Sri Lanka Institute of Information Technology

BSc (Hons) in Information Technology

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ID : **IT23680234**

Assignment 2
Submission Date : 03/04/2026

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Introduction

This assignment demonstrates the implementation of a Business Intelligence solution using a data warehouse. The process includes creating an OLAP cube using SQL Server Analysis Services (SSAS), performing multidimensional analysis using Microsoft Excel, and visualizing data using Power BI. The goal is to support decision-making through data aggregation, analysis, and reporting.

The data warehouse used in this assignment is based on the Online Food Ordering dataset developed in Assignment 1. The dataset contains transactional data related to customers, restaurants, and orders.
The primary fact table is **Fact_Order**, which stores measures such as total amount, delivery fee, and order amount. Dimension tables such as **Dim_Date, Dim_Customer, and Dim_Restaurant** provide descriptive attributes required for analysis.
The dataset supports multidimensional analysis by enabling aggregations across time, customer, and restaurant dimensions.

Data Source Description

Overview of Data Warehouse

The data warehouse was developed using a dimensional model consisting of fact and dimension tables. The fact table stores transactional data such as sales amount, while dimension tables provide descriptive attributes such as customer, product, date, and payment details.

The data warehouse was designed using a **star schema**, as developed in Assignment 1.
The schema consists of:

- **Fact Table**
  - Fact_Order (contains measures such as total_amount, delivery_fee)
- **Dimension Tables**
  - Dim_Date → Year, Month, FullDate
  - Dim_Customer → Customer details (SCD Type 2)
  - Dim_Restaurant → Restaurant details (SCD Type 1)

This structure allows efficient querying and supports analytical operations such as aggregation, filtering, and drill-down.

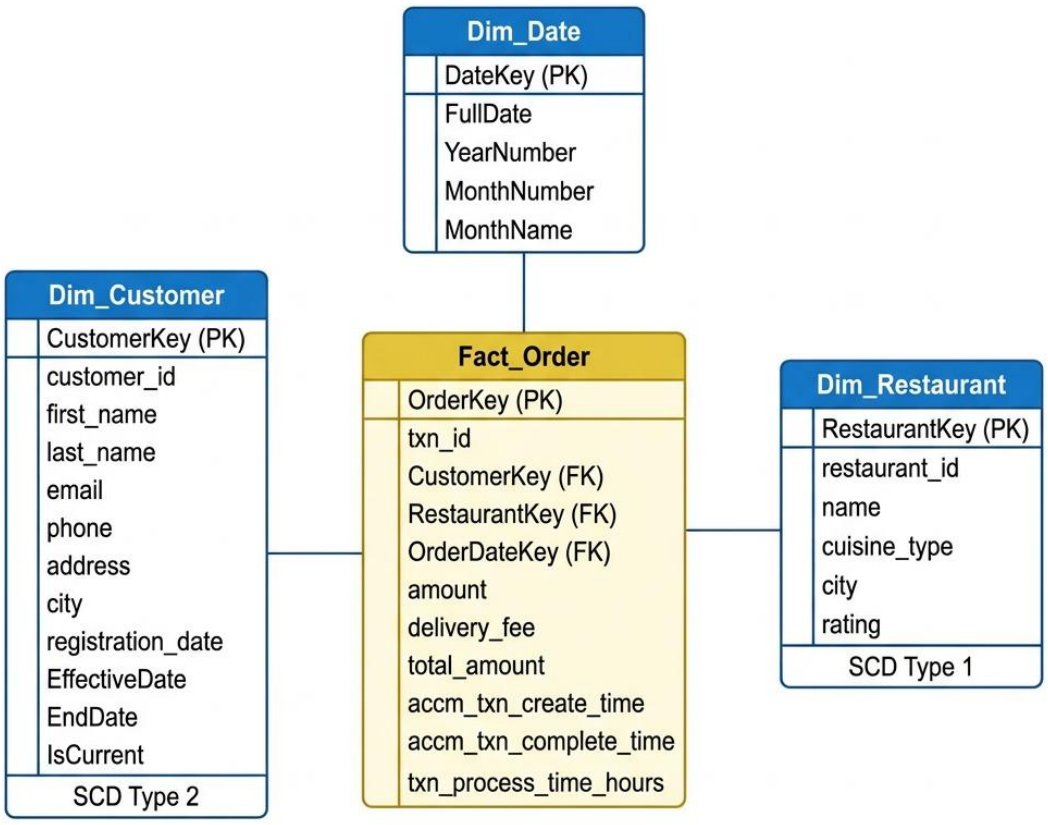
Architecture Overview

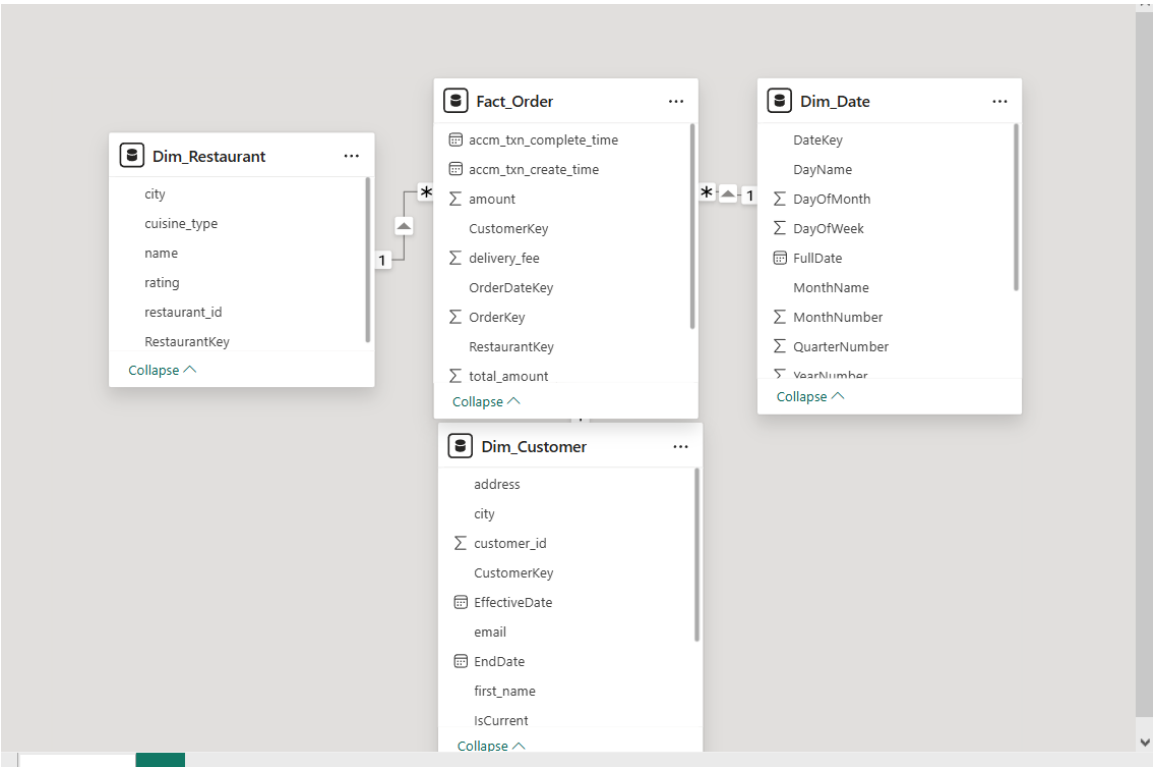
The BI solution follows a layered architecture:

- **Data Source Layer**
  Raw data from OLTP database and flat files (CSV, TXT)
- **ETL Layer (SSIS)**
  Data is extracted, transformed, and loaded into staging and data warehouse
- **Data Warehouse Layer**
  Structured star schema stored in SQL Server
- **BI Layer**
  Analysis and visualization using:
  - SSAS Cube
  - Microsoft Excel (PivotTable)
  - Power BI

This architecture ensures data consistency, scalability, and efficient analysis.

Star Diagram





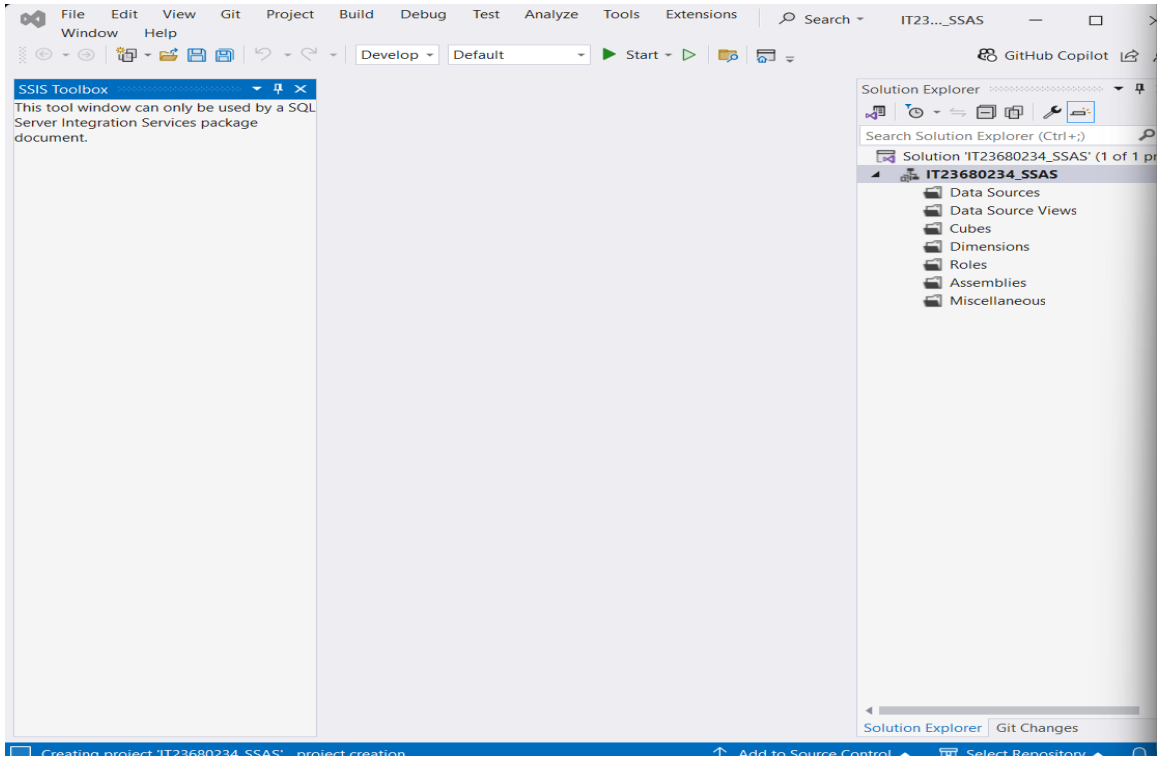
Tables Description

Fact_Order
Stores transactional data such as total_amount, delivery_fee
Linked to all dimensions using surrogate keys
Dim_Date
Provides time-based attributes (Year, Month)
Supports time hierarchy analysis
Dim_Customer
Contains customer details
Implements SCD Type 2 for historical tracking
Dim_Restaurant
Contains restaurant details such as name, city, cuisine type, and rating
Implements SCD Type 1

SSAS Cube Implementation

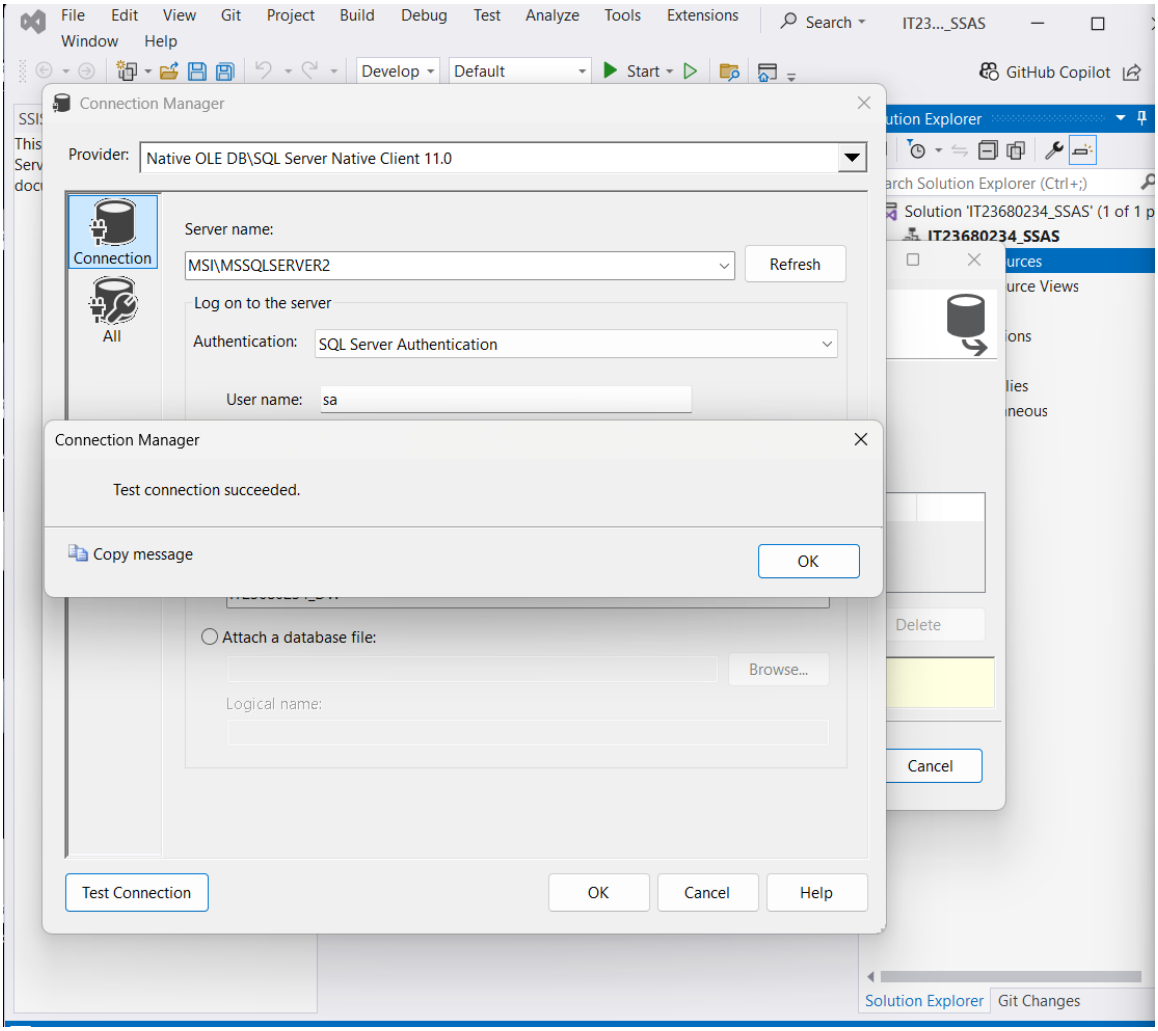
Create SSAS Project

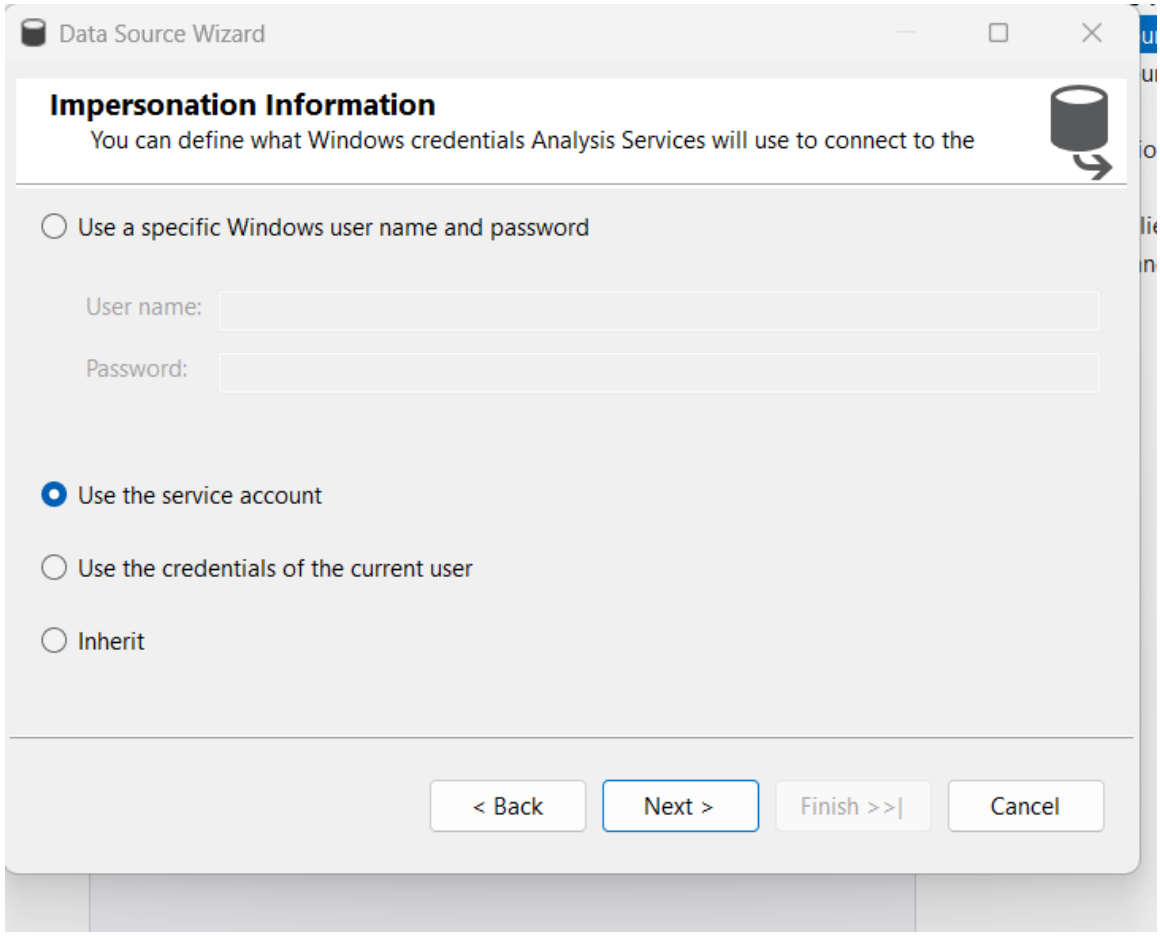
A new Analysis Services Multidimensional Project was created in Visual Studio. The project was connected to the data warehouse database.

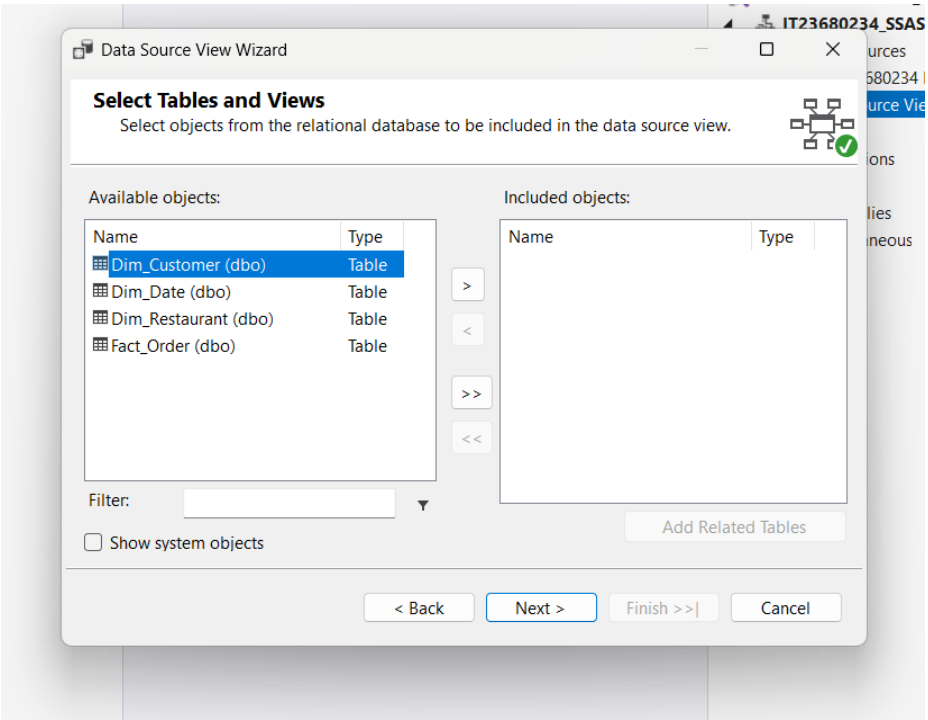


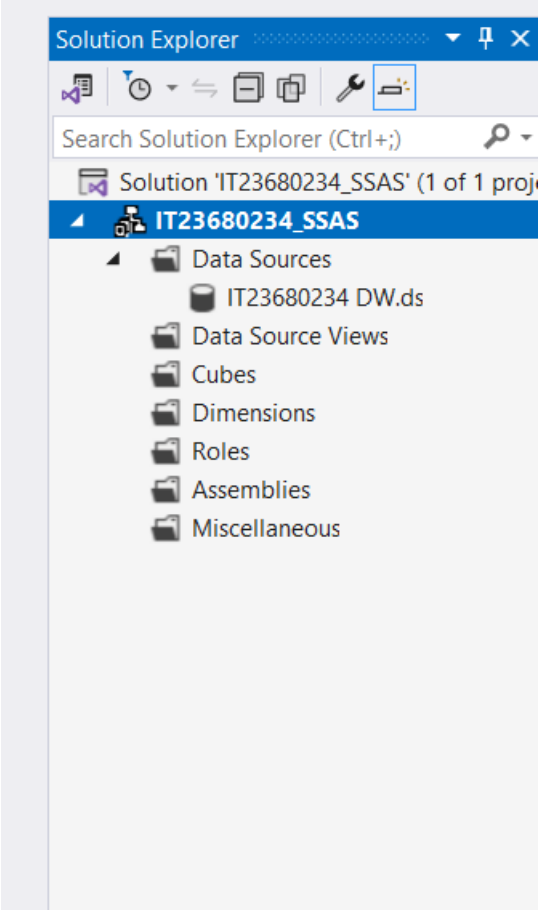
Data Source & Data Source View

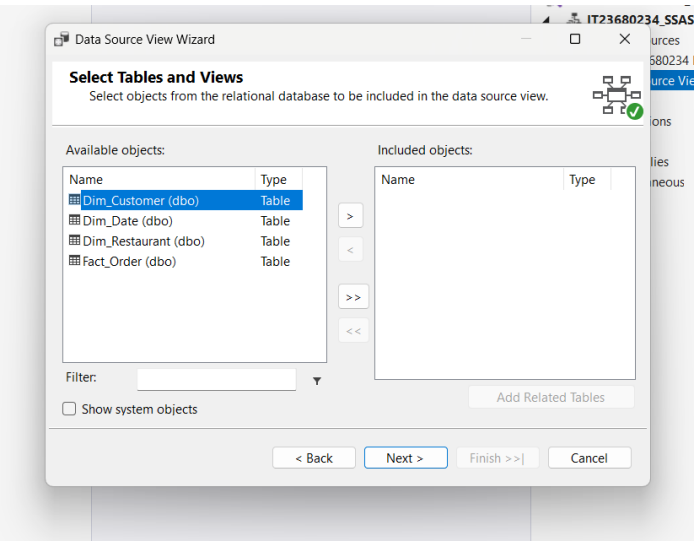
A data source was created by connecting to the data warehouse. A Data Source View (DSV) was created including the fact table and all dimension tables.

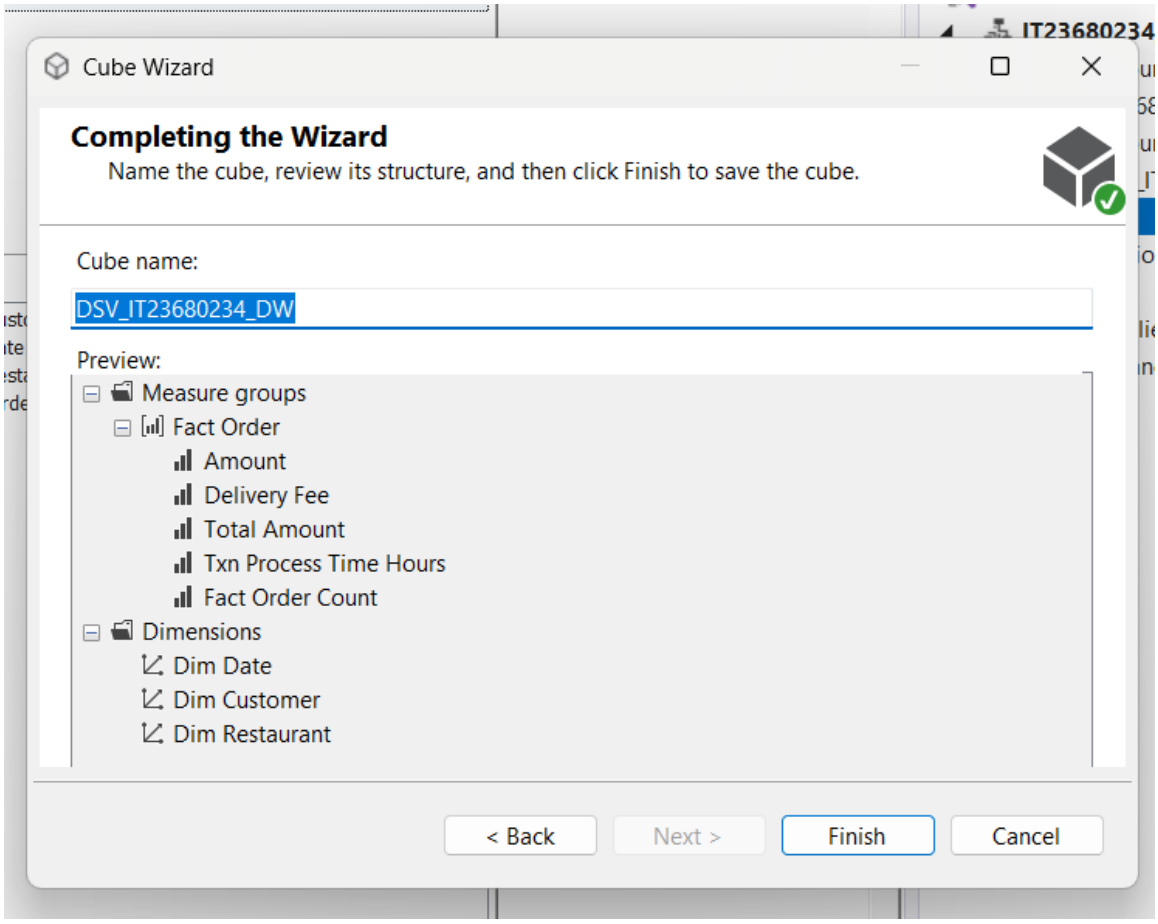



Data Source Wizard
Impersonation Information
You can define what Windows credentials Analysis Services will use to connect to the
Use a specific Windows user name and password
User name:
Password:
Use the service account
Use the credentials of the current user
Inherit
< Back
Next >
Finish >>|
Cancel


Data Source View Wizard
Select Tables and Views
Select objects from the relational database to be included in the data source view.
Available objects:
Name
Type
Dim_Customer (dbo)
Table
Dim_Date (dbo)
Table
Dim_Restaurant (dbo)
Table
Fact_Order (dbo)
Table
Included objects:
Name
Type
Filter:
Show system objects
Add Related Tables
< Back
Next >
Finish >>|
Cancel

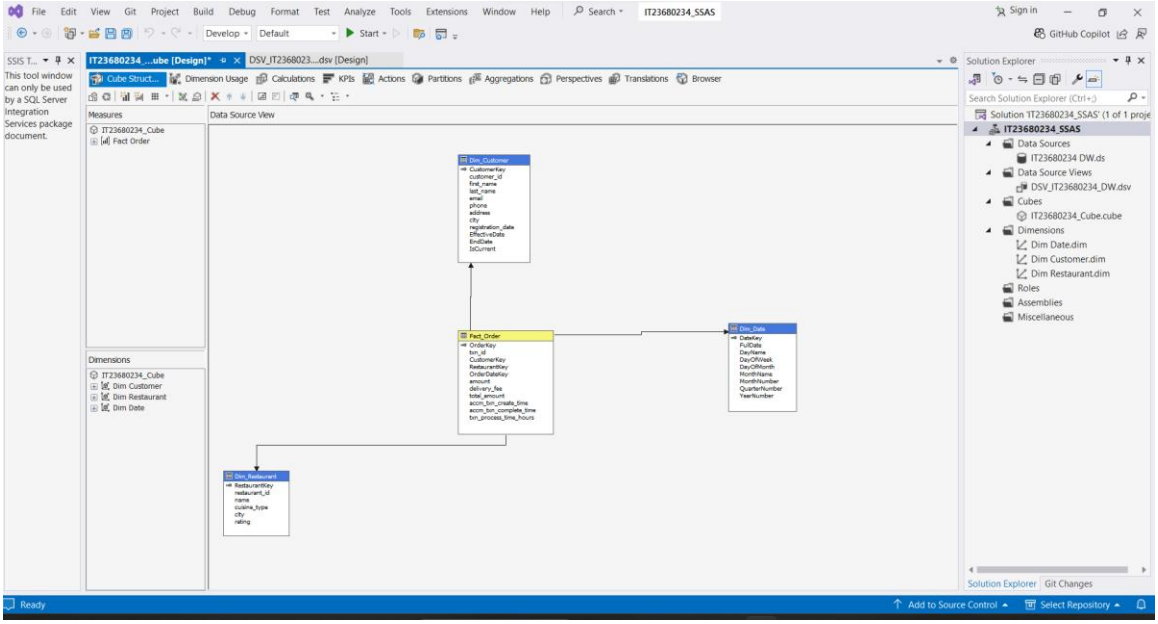

Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'IT23680234_SSAS' (1 of 1 proj
IT23680234_SSAS
Data Sources
IT23680234 DW.ds
Data Source Views
Cubes
Dimensions
Roles
Assemblies
Miscellaneous


Data Source View Wizard
Select Tables and Views
Select objects from the relational database to be included in the data source view.
Available objects:
Name
Type
Dim_Customer (dbo)
Table
Dim_Date (dbo)
Table
Dim_Restaurant (dbo)
Table
Fact_Order (dbo)
Table
Included objects:
Name
Type
Filter:
Show system objects
Add Related Tables
< Back
Next >
Finish >>|
Cancel


Cube Wizard
Completing the Wizard
Name the cube, review its structure, and then click Finish to save the cube.
Cube name:
DSV_IT23680234_DW
Preview:
Measure groups
Fact Order
Amount
Delivery Fee
Total Amount
Txn Process Time Hours
Fact Order Count
Dimensions
Dim Date
Dim Customer
Dim Restaurant
< Back
Next >
Finish
Cancel

Cube Creation

A cube was created using the Cube Wizard. The fact table was selected, and measures such as total amount were automatically generated. Relevant dimensions were added to the cube.

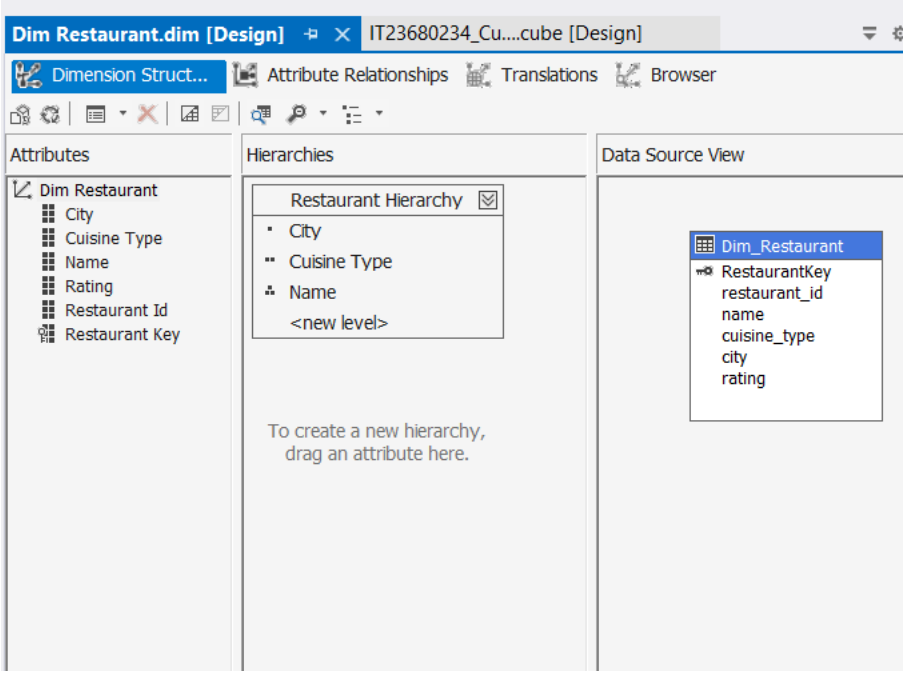


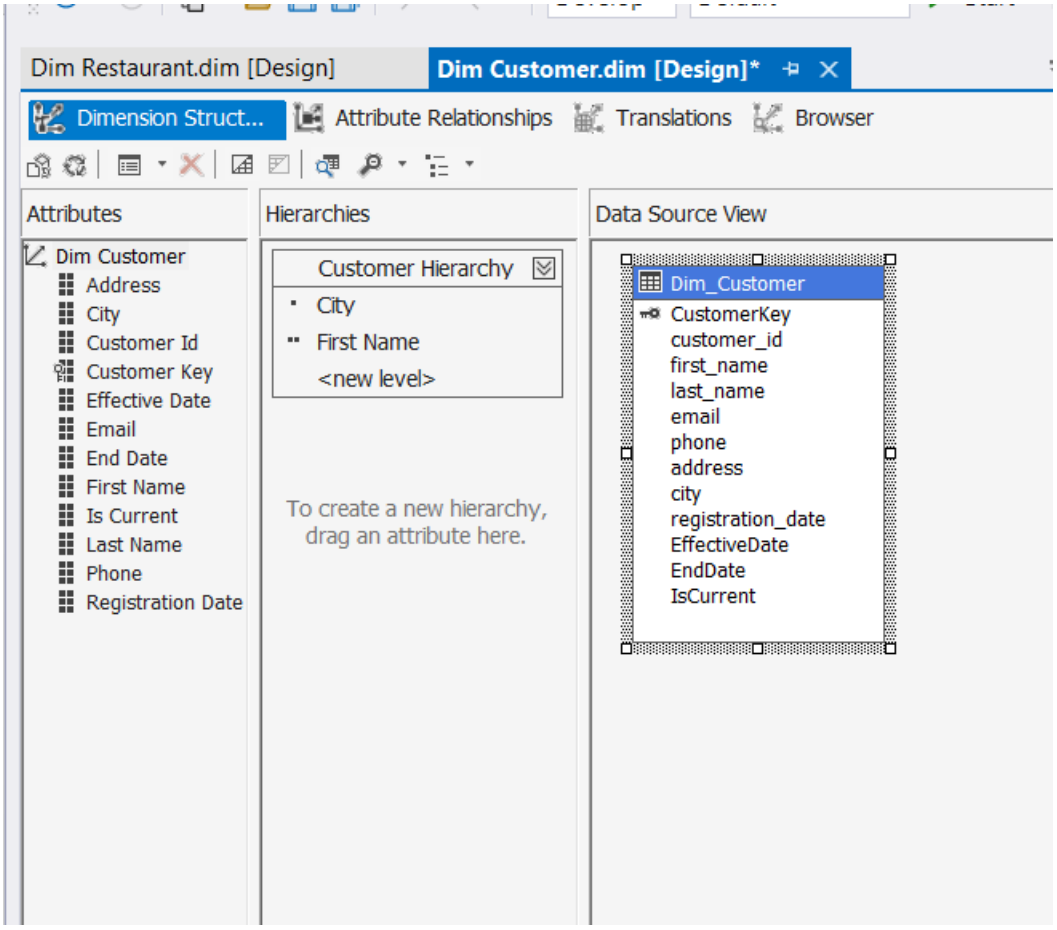
Dimension Creation

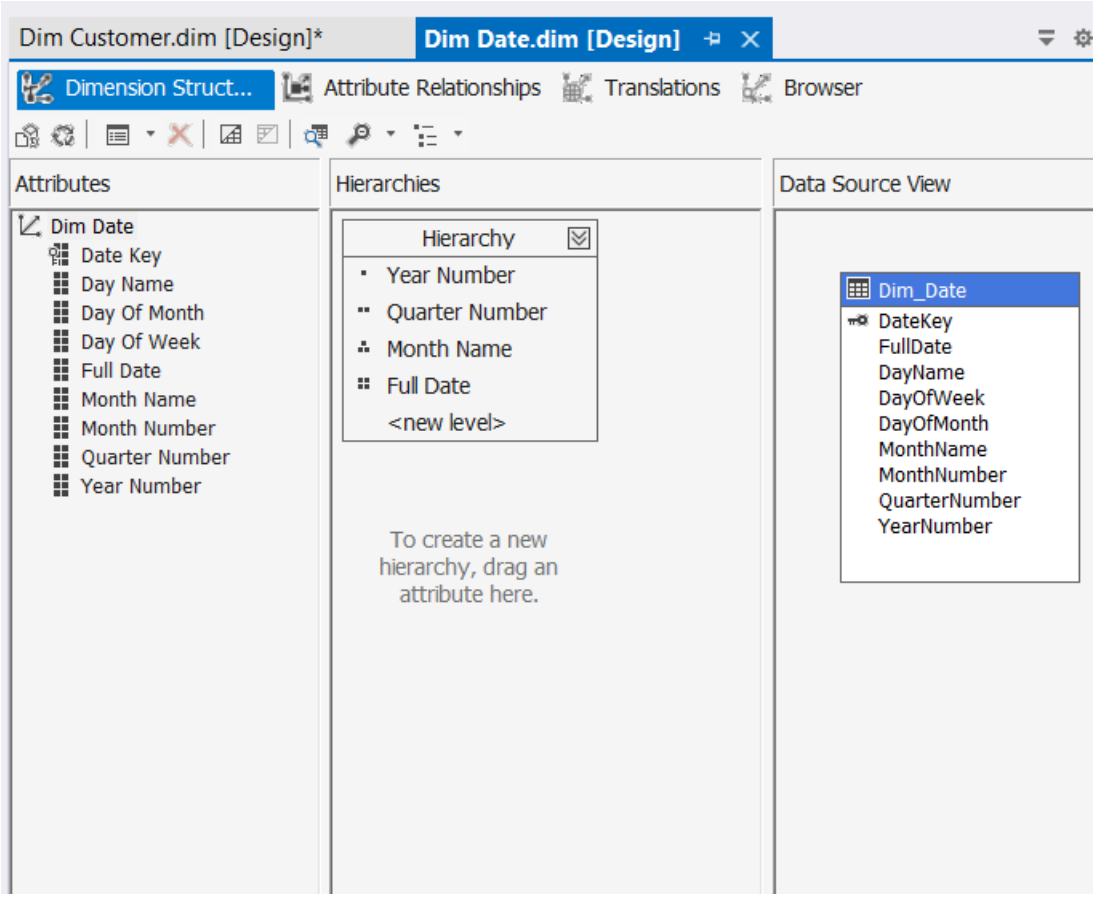
Dimensions were created in SSAS using the dimension tables from the data warehouse:

- Dim_Date
- Dim_Customer
- Dim_Restaurant

Attributes were configured appropriately, and unnecessary technical fields were excluded from analysis.


Dim Restaurant.dim [Design]
IT23680234_Cu....cube [Design]
Dimension Struct...
Attribute Relationships
Translations
Browser
Attributes
Hierarchies
Data Source View
Dim Restaurant
City
Cuisine Type
Name
Rating
Restaurant Id
Restaurant Key
Restaurant Hierarchy
City
Cuisine Type
Name
<new level>
To create a new hierarchy, drag an attribute here.
Dim_Restaurant
RestaurantKey
restaurant_id
name
cuisine_type
city
rating


Dim Restaurant.dim [Design]
Dim Customer.dim [Design]*
Dimension Struct...
Attribute Relationships
Translations
Browser
Attributes
Hierarchies
Data Source View
Dim Customer
Address
City
Customer Id
Customer Key
Effective Date
Email
End Date
First Name
Is Current
Last Name
Phone
Registration Date
Customer Hierarchy
City
First Name
<new level>
To create a new hierarchy, drag an attribute here.
Dim_Customer
CustomerKey
customer_id
first_name
last_name
email
phone
address
city
registration_date
EffectiveDate
EndDate
IsCurrent

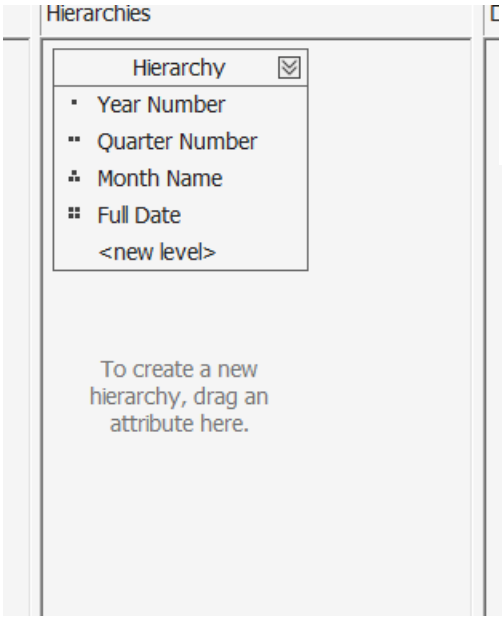

Dim Customer.dim [Design]*
Dim Date.dim [Design]
Dimension Struct...
Attribute Relationships
Translations
Browser
Attributes
Hierarchies
Data Source View
Dim Date
Date Key
Day Name
Day Of Month
Day Of Week
Full Date
Month Name
Month Number
Quarter Number
Year Number
Hierarchy
Year Number
Quarter Number
Month Name
Full Date
<new level>
To create a new hierarchy, drag an attribute here.
Dim_Date
DateKey
FullDate
DayName
DayOfWeek
DayOfMonth
MonthName
MonthNumber
QuarterNumber
YearNumber

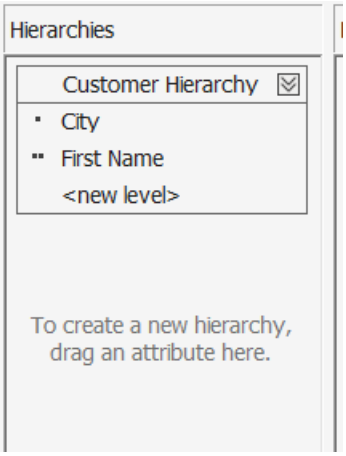
Hierarchy Creation

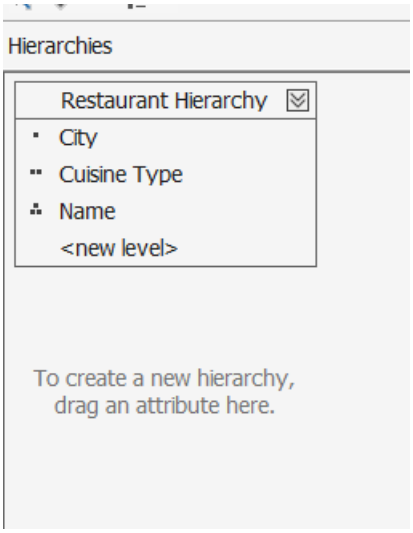
Hierarchies were created to support drill-down analysis:

- **Date Hierarchy**
  - Year → Month
- **Restaurant Hierarchy**
  - City → Restaurant Name

These hierarchies enable users to navigate from summarized data to detailed levels.

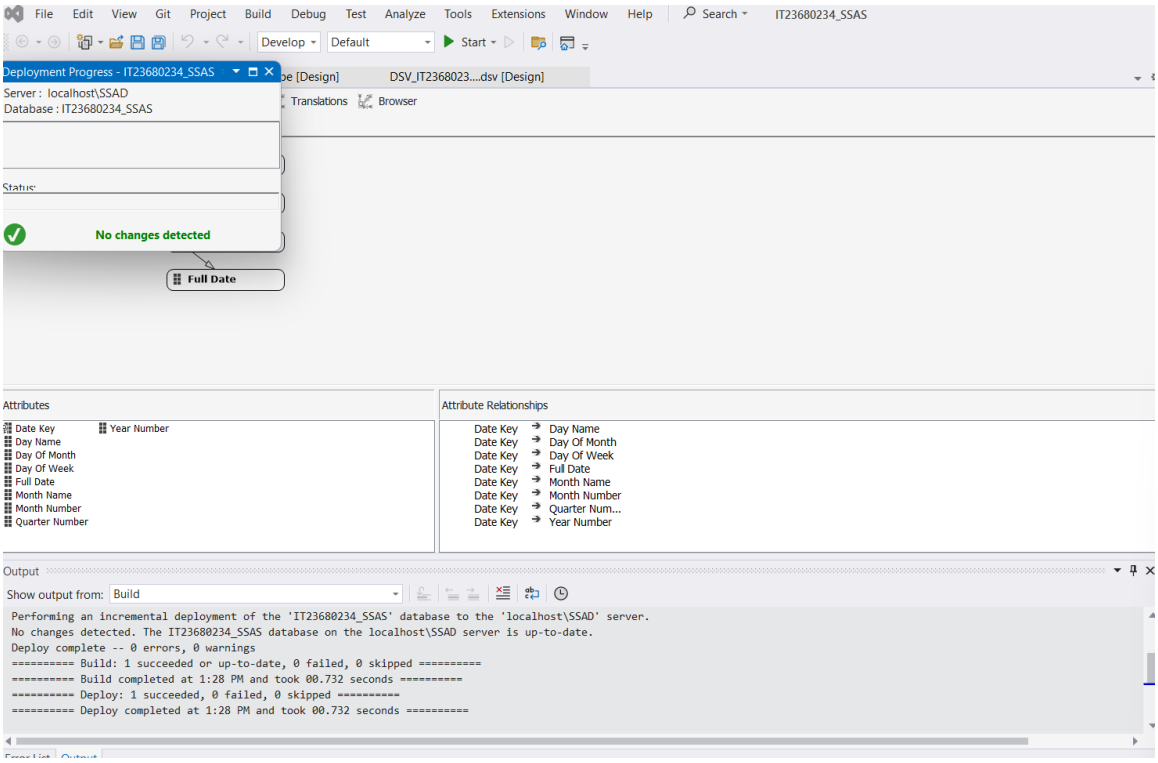






Cube Deployment

The cube was successfully deployed to the Analysis Services server.
Measures such as total_amount and delivery_fee were included, and dimensions were linked correctly to enable multidimensional analysis.



OLAP Operations in Excel

Connect Excel to Cube

Microsoft Excel was connected to the SSAS cube using the Data tab.
Steps followed:

1. Open Excel
2. Go to Data → Get Data → From Database → From Analysis Services
3. Connect to the SSAS server
4. Select the cube
5. Load data into a PivotTable

Pivot Table Creation

A PivotTable was created using the cube data.
Fields were arranged as follows:

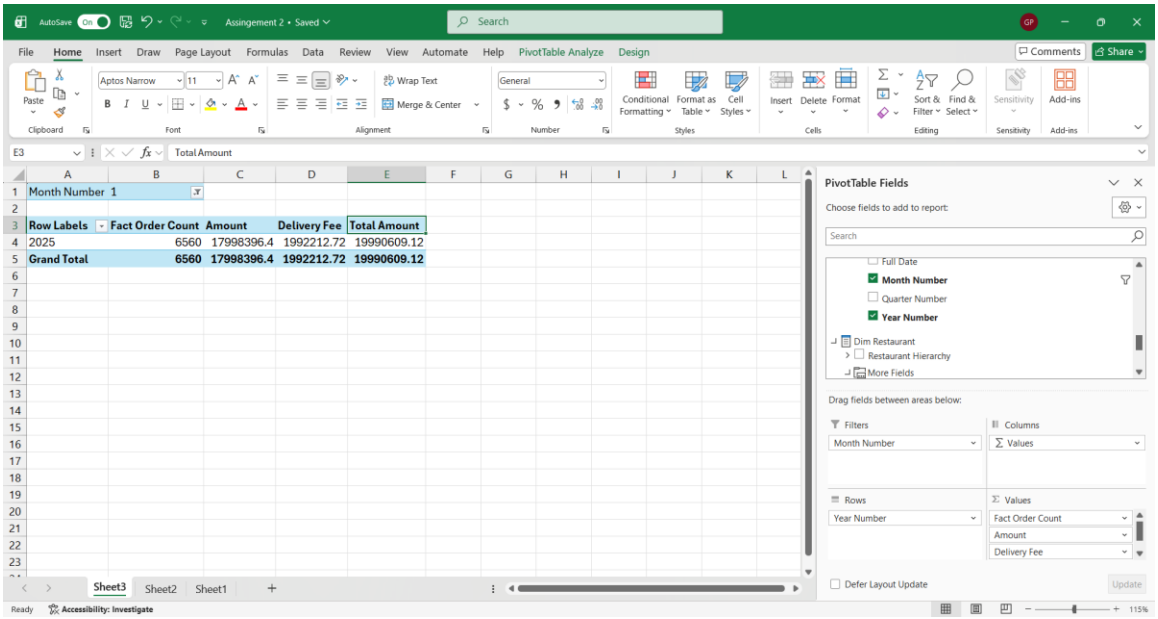
- **Rows**
  - Year → Month
- **Columns**
  - Restaurant City
- **Values**
  - Total Amount (measure)

This provided a structured analytical view of sales data across time and location.

OLAP Operations

Roll-up

Roll-up was performed by aggregating data from Month level to Year level.
The Month attribute was removed, showing only Year-level summaries.



Drill-down

Drill-down was performed by expanding Year into Month using the Date hierarchy.
This allowed viewing detailed monthly data under each year.

Row Labels	Fact Order Count	Amount	Delivery Fee	Total Amount
2025				
1	6560	17998396.4	1992212.72	19990609.12
10	6656	18544836.4	1955502.48	20500338.88
11	6456	17225728.08	1923476.72	19149204.8
12	6936	19119013.12	2071016.16	21190029.28
2	5960	16433833.28	1799235.76	18233069.04
3	7056	19283576.64	2084087.04	21367663.68
4	6136	17119502.24	1792399.6	18911901.84
5	7280	19920948.16	2211127.36	22132075.52
6	6480	17710585.92	1974244.4	19684830.32
7	6720	18705021.68	2003171.84	20708193.52
8	6992	18697881.28	2102292.96	20800174.24
9	6768	18393687.52	1980808.64	20374496.16
Grand Total	80000	219153010.7	23889575.68	243042586.4

Slice

Slice operation was performed by applying a filter using a single dimension such as Cuisine Type.
This enabled analysis of data for a specific category.

E3 | Total Amount

	A	B	C	D	E	F	G
1	Month Number	1					
2							
3	**Row Labels**	**Fact Order Count**	**Amount**	**Delivery Fee**	**Total Amount**		
4	2025	6560	17998396.4	1992212.72	19990609.12		
5	**Grand Total**	**6560**	**17998396.4**	**1992212.72**	**19990609.12**		
6							
7							
8							
9							
10							
11							
12							

Dice

Dice operation was performed by applying multiple filters such as City, Cuisine Type, and Year.
This allowed analysis of a specific subset of data.

B3 | North Indian

	A	B	C	D	E	F	G
1	Month Number	1					
2	City	Delhi					
3	Cuisine Type	North Indian					
4							
5	**Row Labels**	**Fact Order Count**	**Amount**	**Delivery Fee**	**Total Amount**		
6	2025	294	890110.38	85218.54	975328.92		
7	**Grand Total**	**294**	**890110.38**	**85218.54**	**975328.92**		
8							
9							
10							
11							
12							

Pivot

Pivot operation was performed by swapping rows and columns.
For example:

- Rows changed from Year to City
- Columns changed from City to Year

This provided a different perspective of the same data.

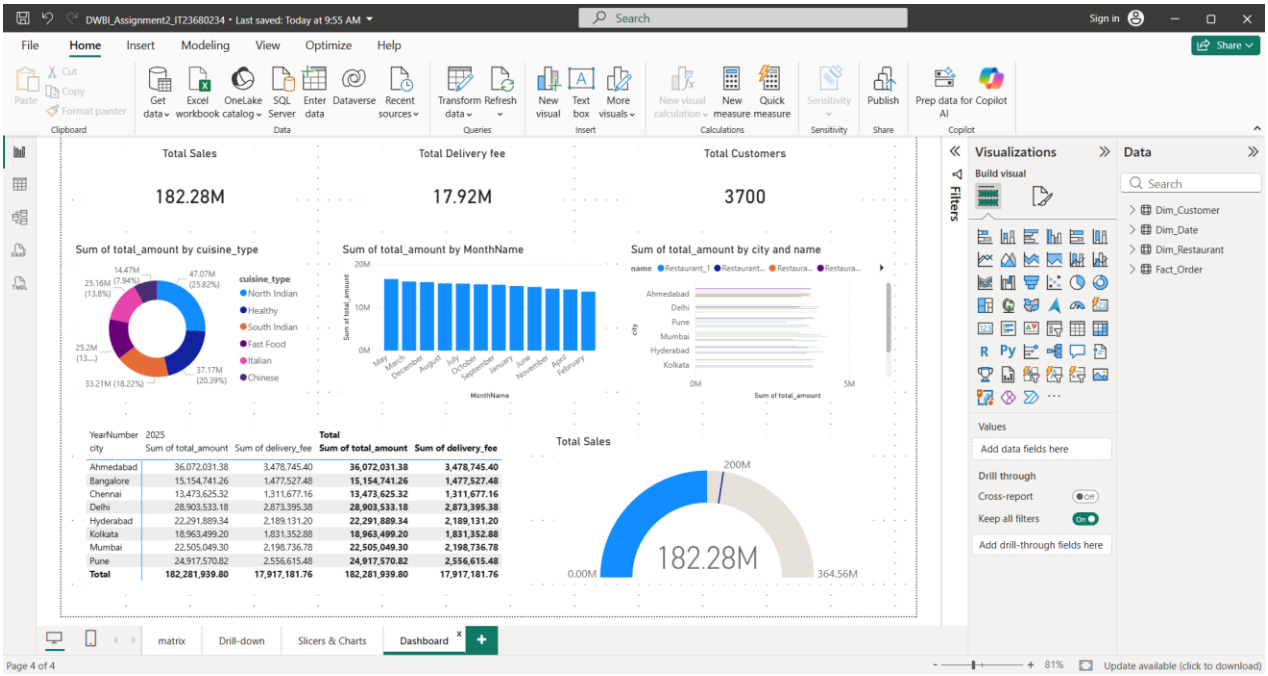
	Column Labels			
	Fact Order Count	Amount	Delivery Fee	Total Amount
Row Labels	2025	2025	2025	2025
Ahmedabad	11772	32593285.98	3478745.4	36072031.38
Bangalore	4902	13677213.78	1477527.48	15154741.26
Chennai	4314	12161948.16	1311677.16	13473625.32
Delhi	9624	26030137.8	2873395.38	28903533.18
Hyderabad	7350	20102758.14	2189131.2	22291889.34
Kolkata	6126	17132146.32	1831352.88	18963499.2
Mumbai	7416	20306312.52	2198736.78	22505049.3
Pune	8496	22360955.34	2556615.48	24917570.82
Grand Total	**60000**	**164364758**	**17917181.76**	**182281939.8**

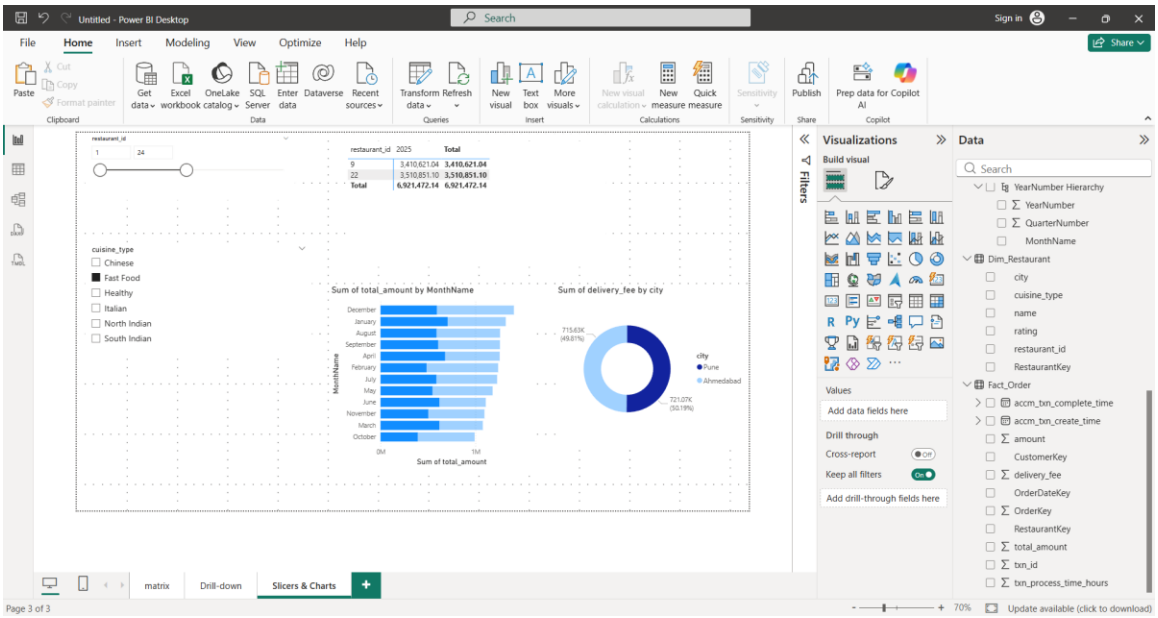
Power BI Reports

Power BI was used to create interactive dashboards based on the cube data.
The dashboard includes:

- KPI Cards (Total Sales, Delivery Fee)
- Bar Charts (Sales by Category/City)
- Pie/Donut Charts (Payment Type, Year distribution)
- Matrix Visual (detailed breakdown)

Interactive features such as slicers were added to filter data dynamically.
These reports provide a visually rich and interactive way to analyze business performance.




Untitled - Power BI Desktop
Search
Sign in
File
Home
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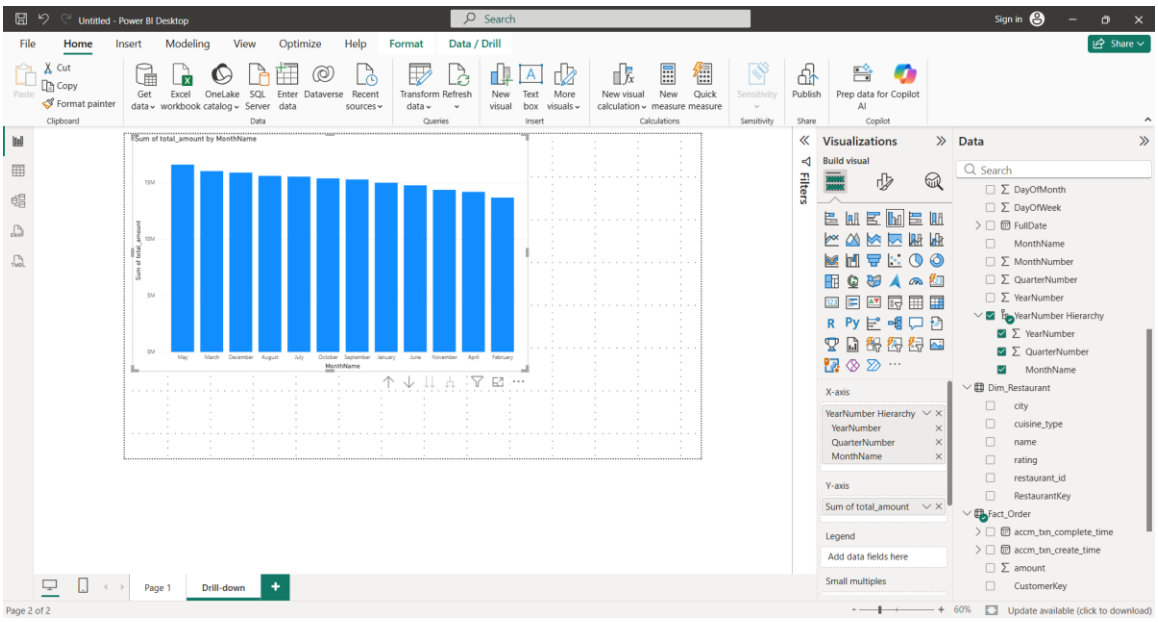
Modeling
View
Optimize
Help
Share
Paste
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for Copilot AI
Copilot
restaurant_id
1
24
restaurant_id 2025 Total
9 3,410,621.04 3,410,621.04
22 3,510,851.10 3,510,851.10
Total 6,921,472.14 6,921,472.14
cuisine_type
Chinese
Fast Food
Healthy
Italian
North Indian
South Indian
Sum of total_amount by MonthName
December
January
August
September
April
February
July
May
June
November
March
October
MonthName
0M
1M
Sum of total_amount
Sum of delivery_fee by city
715.63K (49.81%)
721.07K (50.19%)
city
Pune
Ahmedabad
matrix
Drill-down
Slicers & Charts
Page 3 of 3
70%
Update available (click to download)
Filters
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Off
Keep all filters
On
Add drill-through fields here
Data
Search
YearNumber Hierarchy
YearNumber
QuarterNumber
MonthName
Dim_Restaurant
city
cuisine_type
name
rating
restaurant_id
RestaurantKey
Fact_Order
accm_txn_complete_time
accm_txn_create_time
amount
CustomerKey
delivery_fee
OrderDateKey
OrderKey
RestaurantKey
total_amount
txn_id
txn_process_time_hours


Untitled - Power BI Desktop
Search
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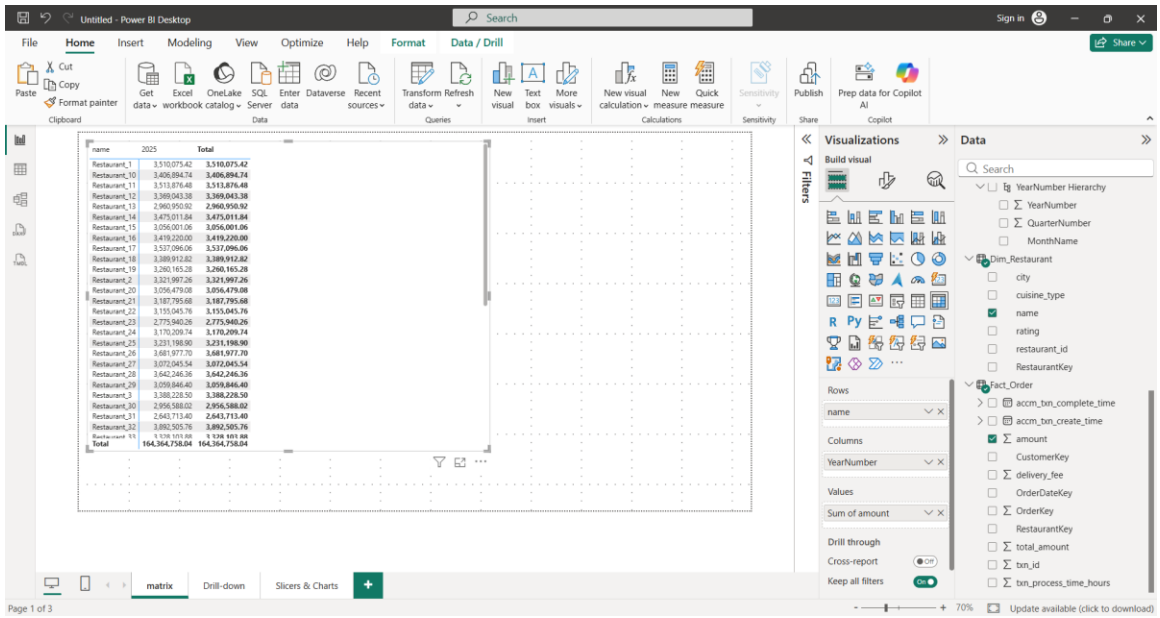
Copy
Paste
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for Copilot AI
Copilot
Sum of total_amount by MonthName
15M
10M
5M
0M
Sum of total_amount
May
March
December
August
July
October
September
January
June
November
April
February
MonthName
Page 1
Drill-down
Page 2 of 2
60%
Update available (click to download)
Filters
Visualizations
Build visual
X-axis
YearNumber Hierarchy
YearNumber
QuarterNumber
MonthName
Y-axis
Sum of total_amount
Legend
Add data fields here
Small multiples
Data
Search
DayOfMonth
DayOfWeek
FullDate
MonthName
MonthNumber
QuarterNumber
YearNumber
YearNumber Hierarchy
YearNumber
QuarterNumber
MonthName
Dim_Restaurant
city
cuisine_type
name
rating
restaurant_id
RestaurantKey
Fact_Order
accm_txn_complete_time
accm_txn_create_time
amount
CustomerKey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Format
Data / Drill
Share
Paste
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for AI
Copilot
name 2025 Total
Restaurant_1 3,510,075.42 3,510,075.42
Restaurant_10 3,406,894.74 3,406,894.74
Restaurant_11 3,513,876.48 3,513,876.48
Restaurant_12 3,369,043.38 3,369,043.38
Restaurant_13 2,960,950.92 2,960,950.92
Restaurant_14 3,475,011.84 3,475,011.84
Restaurant_15 3,056,001.06 3,056,001.06
Restaurant_16 3,419,220.00 3,419,220.00
Restaurant_17 3,537,096.06 3,537,096.06
Restaurant_18 3,389,912.82 3,389,912.82
Restaurant_19 3,260,165.28 3,260,165.28
Restaurant_2 3,321,997.26 3,321,997.26
Restaurant_20 3,056,479.08 3,056,479.08
Restaurant_21 3,187,795.68 3,187,795.68
Restaurant_22 3,155,045.76 3,155,045.76
Restaurant_23 2,775,940.26 2,775,940.26
Restaurant_24 3,170,209.74 3,170,209.74
Restaurant_25 3,231,198.90 3,231,198.90
Restaurant_26 3,681,977.70 3,681,977.70
Restaurant_27 3,072,045.54 3,072,045.54
Restaurant_28 3,642,246.36 3,642,246.36
Restaurant_29 3,059,846.40 3,059,846.40
Restaurant_3 3,388,228.50 3,388,228.50
Restaurant_30 2,956,588.02 2,956,588.02
Restaurant_31 2,643,713.40 2,643,713.40
Restaurant_32 3,892,505.76 3,892,505.76
Total 164,364,758.04 164,364,758.04
Visualizations
Build visual
Filters
Rows
name
Columns
YearNumber
Values
Sum of amount
Drill through
Cross-report
Off
Keep all filters
On
Data
Search
YearNumber Hierarchy
YearNumber
QuarterNumber
MonthName
Dim_Restaurant
city
cuisine_type
name
rating
restaurant_id
RestaurantKey
Fact_Order
accm_txn_complete_time
accm_txn_create_time
amount
CustomerKey
delivery_fee
OrderDateKey
OrderKey
RestaurantKey
total_amount
txn_id
txn_process_time_hours
matrix
Drill-down
Slicers & Charts
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